

Surname : SOLUTION GUIDE Initials :

Student number:

STICK BARCODE STICKER HERE

$\checkmark \equiv \text{full mark}$
 $\checkmark^{\wedge} \equiv \frac{1}{2} \text{ mark}$



FSK 116
Semester Test 1
Department of Physics, University of Pretoria

Date: 2025-03-26	Time: 3.0 hours	Total: 75
Examiner: RQ Odendaal		Moderator: Prof JM Nel

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Test register: Tear carefully on the line above and hand to the invigilator.
All exam regulations are applicable to this test as indicated in point 8 on the next page.

FSK 116: Semester Test 1 **2025-03-26**

Surname		Initials	
Student number			
Venue			
Signature			

STICK BARCODE STICKER HERE

Please read the following carefully.

1. Answer all the questions in the space provided on the question paper.
2. No pencil work will be marked. Do not use "tippex".
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues.
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.**
6. **Units must be used when numbers are substituted else marks will be subtracted.**
7. The use of the cheat sheet is only allowed subject to the rules stated on it.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner.

Constants and formulas

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$;
 $1 \text{ atm} = 101.3 \text{ kPa}$; $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 299\,792.5 \text{ km/s}$; $V_{\text{sphere}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$;
 $1 \text{ litre} = 1000 \text{ cm}^3$; $M_{\text{Earth}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = 6.37 \times 10^6 \text{ m}$; $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$

1 Question 1

[7]

Andy Warhol was a famous artist. He once said that "everyone will be world-famous for 15 minutes".

- (a) If we define a new unit of 1 warhol = 15 minutes, how long is 1 megawarhol in years? Show the conversion factors in your calculation.

$$\begin{aligned}
 1 \text{ warhol} &= 15 \text{ min} \\
 1 \text{ Megawarhol} &= \frac{10^6 \text{ warhol}}{1} \times \frac{15 \text{ min}}{1 \text{ warhol}} \times \dots \times \dots \times \dots \\
 &= \dots \\
 &= 28.54 \text{ years} \checkmark
 \end{aligned}$$

3 x 1/2 marks (4)
other factors

4

- (b) Photons move at a constant speed of 300 000 km/s. How far (in meters) would a photon move in 1 Megawhol? Show the conversion factors in your calculation. Use the given data in your calculation.

3

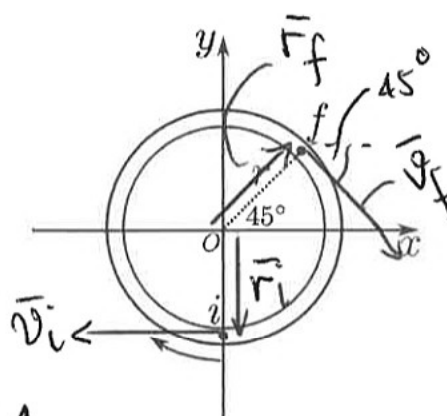
$$\begin{aligned} \text{distance} &= ct \\ &= \frac{300\,000\text{ km}}{1\text{ s}} \times \frac{10^3\text{ m}}{1\text{ km}} \times \frac{10^6\text{ whol}}{1} \times \dots \\ &= \dots \\ &= 2.70 \times 10^{17}\text{ m} \checkmark \end{aligned}$$

(3) 2 x 1/2 marks

2 Question 2

[9]

An athlete runs from position i , and follows the circular track clockwise up to the point f as shown. If $r = 16\text{ m}$, and the athlete had a speed of $v_i = 4\text{ m/s}$ initially, and a speed of $v_f = 8\text{ m/s}$ finally, calculate the following:



- (a) Her displacement from point i to point f , using the origin as indicated in the figure.

(3)

3

$$\begin{aligned} \vec{r}_i &= -(16\text{ m})\hat{j} \\ \vec{r}_f &= (r \cos 45^\circ)\hat{i} + (r \sin 45^\circ)\hat{j} \\ &= \dots \text{ calculate value.} \end{aligned}$$

$$\Delta \vec{r} = \vec{r}_f - \vec{r}_i = \dots = (11.3\text{ m})\hat{i} + (27.3\text{ m})\hat{j}$$

Displacement is a vector!

- (b) What is the **average** acceleration from point i to point f of the athlete if it takes 10.2 s to move between the two points? (6)

6

$$\vec{v}_i = (-4 \text{ m/s}) \hat{i} \quad \checkmark \quad \text{See sketch}$$

$$\vec{v}_f = (v_f \cos 45^\circ) \hat{i} - (v_f \sin 45^\circ) \hat{j} \quad \checkmark$$

$$= \dots \dots \text{numerical answer} \quad \checkmark$$

$$\Delta \vec{v} = \vec{v}_f - \vec{v}_i = \dots = (9.7 \text{ m/s}) \hat{i} - (5.7 \text{ m/s}) \hat{j} \quad \checkmark$$

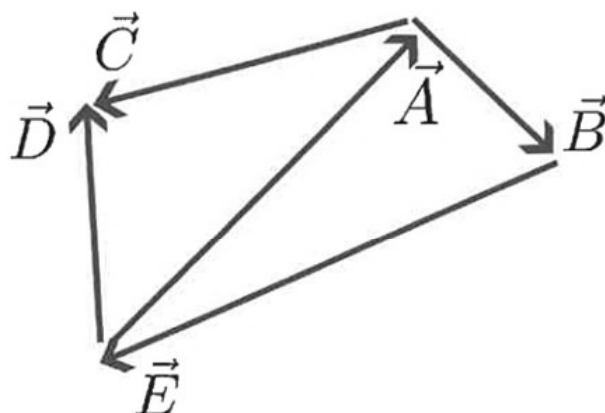
vector! $\rightarrow \vec{a}_{\text{avg}} = \frac{\Delta \vec{v}}{\Delta t} = (0.95 \text{ m/s}^2) \hat{i} - (0.56 \text{ m/s}^2) \hat{j} \quad \checkmark.$

3 Question 3

[5]

5

Refer to the figure below. Express the following vectors in terms of vectors $\vec{A}$ and $\vec{B}$:



(a) $\vec{E}$

$$-\vec{E} = \vec{A} + \vec{B} \quad \checkmark \quad \text{or} \quad \vec{E} = -(\vec{A} + \vec{B}) = -\vec{A} - \vec{B}$$

(b) $\vec{D} - \vec{C}$

$$\vec{D} + (-\vec{C}) = \vec{A} \quad \checkmark$$

(c) $\vec{E} + \vec{D} - \vec{C}$

$$\vec{E} + (\vec{D} - \vec{C}) = \vec{E} + \vec{A} = -\vec{B} \quad \checkmark \checkmark$$

4 Question 4

[8]

A bird flies in the xy -plane with a velocity vector given by

$$\vec{v}(t) = (2 \text{ m} \cdot \text{s}^{-1} - 3 \text{ m} \cdot \text{s}^{-3} t^2) \hat{i} + (5 \text{ m} \cdot \text{s}^{-2} t) \hat{j},$$

where the positive y -direction is vertically upward. At $t = 0$, the bird is at the origin.

(a) Calculate the position and the acceleration vectors of the bird as functions of time.

(5)

5

To find $\vec{r}$ we need to integrate $\vec{v}(t)$; it gives $\Delta \vec{r} = \vec{r}(t) - \vec{r}(t=0)$

$$= [(2 \text{ m/s})t - (1 \text{ m/s}^3)t^3] \hat{i} + [(2.5 \text{ m/s}^2)t^2] \hat{j} \quad \text{--- ①}$$

We differentiate to find $\vec{a}(t) = \frac{d\vec{v}}{dt}$:

$$\vec{a}(t) = [(-6 \text{ m/s}^3)t] \hat{i} + [(5 \text{ m/s}^2)] \hat{j} \quad \checkmark$$

(b) What is the bird's altitude as it flies over $x = 0$ for the first time after $t = 0$?

(3)

3

Use ① in (a) to solve when $(t) \quad x=0$:

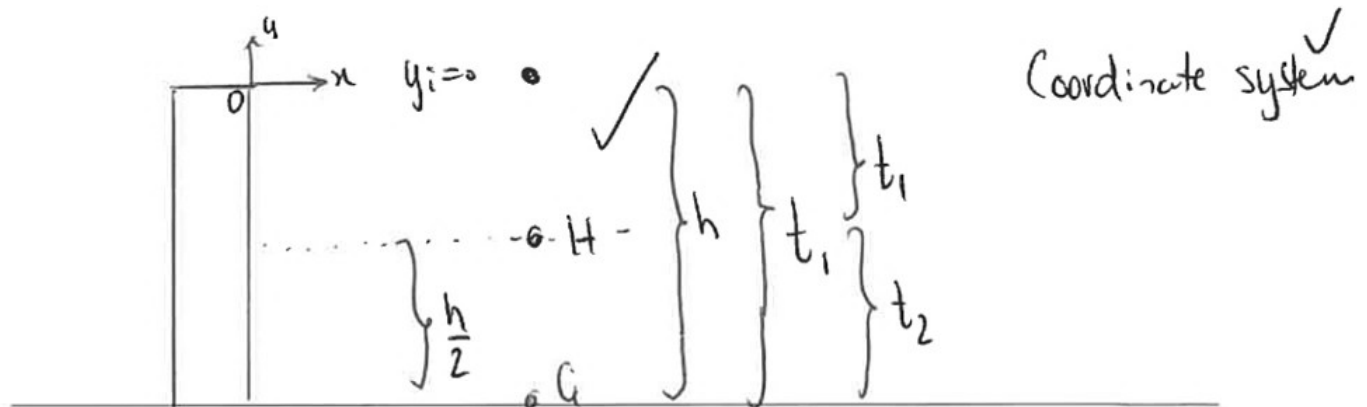
$$\Rightarrow [(2 \text{ m/s})t - (1 \text{ m/s}^3)t^3] = 0 \quad \checkmark$$

$$t = \sqrt{2} \text{ s} \quad \checkmark$$

To find altitude substitute the time into y -component of ①:
 $y = 5 \text{ m} \quad \checkmark$

A rock dropped from the top of a building takes 0.50 s to fall the last 50% of the distance from the top of the building to the ground. How tall is the building?

Carefully sketch your problem in the space below. Label all important points, and use them to identify your symbols. First find a suitable symbolic solution, before substituting any numbers in the solution.



Falling from the top to the ground, point G:

$$y_G = y_i + v_{yi} t_a - \frac{1}{2} g t_a^2$$

$$t_a = \sqrt{\frac{2h}{g}} \quad \text{but } t_a = t_1 + t_2 = \sqrt{\frac{2h}{g}} \quad \text{--- (1)}$$

Falling from the top of the building to point H:

Solve for time t_1

$$t_1 = \sqrt{\frac{h}{g}} \quad \text{--- (2)}$$

$$\text{From (1) : } t_2 = \sqrt{\frac{2h}{g}} - t_1 = \dots = (\sqrt{2} - 1) \sqrt{\frac{h}{g}} \quad \text{--- (3)}$$

Solve for h from (3): $h \approx 14.3 \text{ m}$ ✓

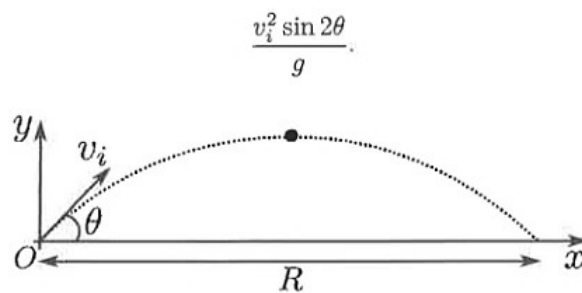
6

Question 6

[7]

7

The figure shows a projectile being launched from the origin O of the coordinate system. The projectile is launched at an angle of θ with an initial speed v_i . If the position of the projectile is given by $\vec{r} = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$ at any time t , and the velocity of it is given as $\vec{v} = \vec{v}_i + \vec{a} t$ for any time t , show that the range, R , of the projectile is given by



See textbook page 76 for the derivation or the class notes.

Derivations are marked strictly. You need to start with the eq.'s given.

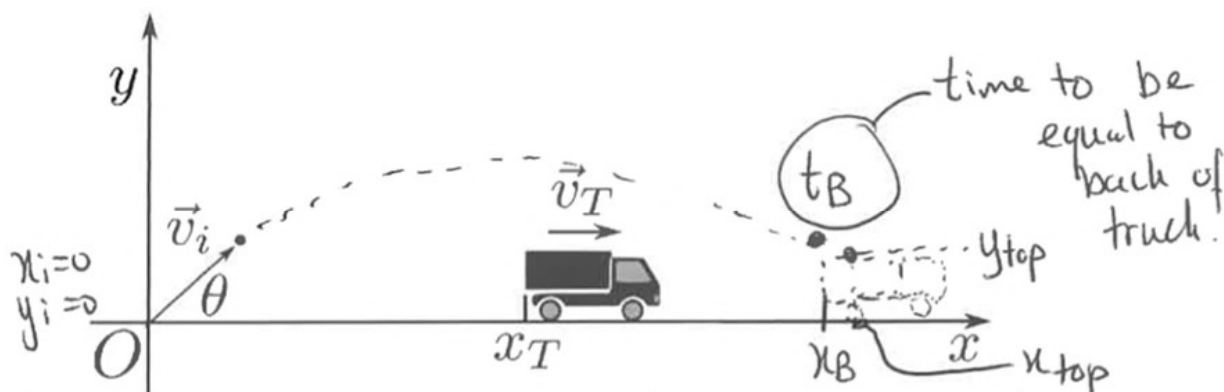
Notation must be correct to receive full marks.

A projectile is fired with an initial speed $v_i = 35 \text{ m/s}$ at an angle $\theta = 23^\circ$. A truck is moving along the x -axis with a constant speed of $v_T = 15 \text{ m/s}$. At the moment the projectile is fired, the back of the truck is at $x_T = 45 \text{ m}$.

Will the projectile hit the back of the truck?

Now, find the coordinates (x_H, y_H) where the projectile hits the truck if the truck is 2.0 m tall. You may assume that the projectile is launched very close to the ground.

Modify the sketch below and clearly indicate all the positions relevant to the problem.



Truck's motion in x -direction is given by

$$x = x_T + v_T t \quad - (1)$$

Projectile and back of truck is at x_B when

$$\underbrace{x_i + v_{xi} t_B}_{\text{proj.'s } x\text{-motion}} = \underbrace{x_T + v_T t_B}_{\text{truck's } x\text{-motion}}$$

$$0 + v_i \cos \theta t_B = x_T + v_T t_B \quad \checkmark$$

$$\text{Solve for } t_B = 2.614 \text{ s} \quad \checkmark$$

Check y -pos. of proj. $\Rightarrow$ eq $\checkmark$ to solve

$$\Rightarrow y_B = 2.25 \text{ m (above the truck)} \quad \checkmark$$

When is projectile on the roof?

$$y_{\text{top}} = y_i + v_i \sin \theta t_{\text{top}} - \frac{1}{2} g t_{\text{top}}^2 \quad \checkmark$$

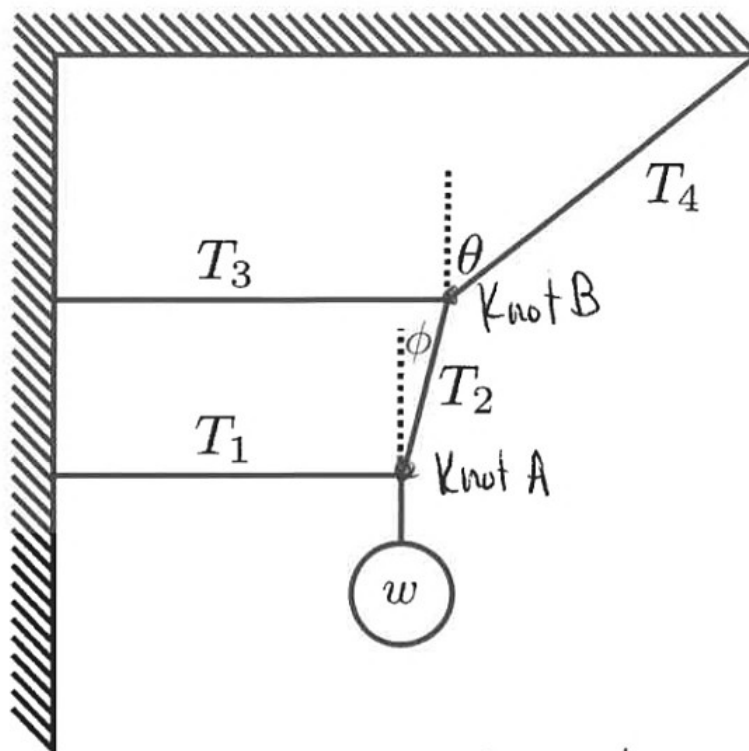
$$\text{Solving for } t_{\text{top}} = 2.635 \text{ s} \quad \checkmark$$

Truck is hit after $2.635 \text{ s} - 2.614 \text{ s} = 0.021 \text{ s} \quad \checkmark$

$$\begin{aligned} x\text{-position where truck is hit: } x_{\text{top}} &= [v_i \cos \theta - v_T] (t_{\text{top}} - t_B) \\ &= \dots = 36 \text{ cm} \quad \checkmark \end{aligned}$$

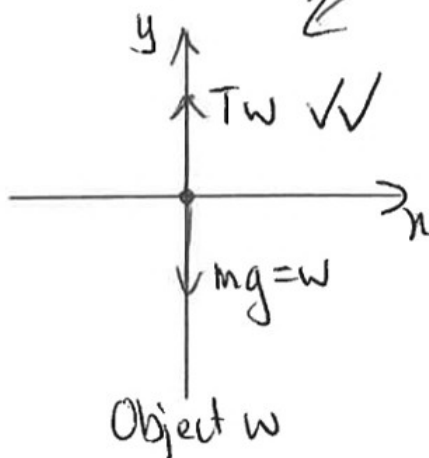
Truck is hit on top of roof, 36 cm from the back.

The weight w in the figure is 80 N. Find the tensions T_1 , T_2 , T_3 and T_4 in the ropes. The angles are $\theta = 55^\circ$ and $\phi = 25^\circ$.

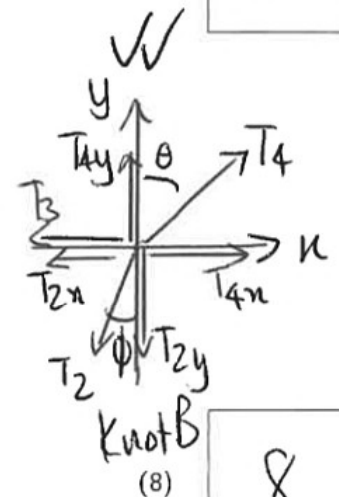
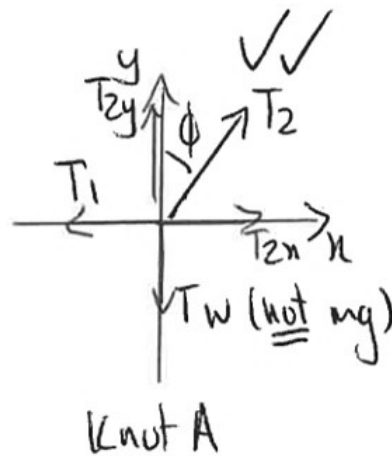


Sketch your free-body diagrams (FBDs).

3 objects



Coordinate systems must be shown



Solve for each tension: One mark is given for each eq. and one mark for the correct numerical value. (T_1 to T_4)

$$T_w = 80 \text{ N}, T_2 = 88.3 \text{ N}, T_1 = 37.3 \text{ N}, T_4 = 139.5 \text{ N}$$

$$\text{and } T_3 = 76.9 \text{ N}$$

9 Question 9

[6]

A fly of mass 0.2 g sits 12 cm from the center of a vinyl record revolving at $33\frac{1}{3}$ rpm (revolutions per minute).

(a) What is the magnitude of the centripetal force on the fly?

(4)

4

Period $T = \frac{2\pi r}{v}$ and $\omega = \frac{2\pi}{T}$

$$v = \omega r = (33.33 \frac{\text{rev}}{\text{min}}) \left(\frac{2\pi}{\text{rev}} \right) \left(\frac{1 \text{ min}}{60 \text{ s}} \right) \times (12 \times 10^{-2} \text{ m})$$

$$= 0.419 \text{ m/s}$$

$$F_c = m \frac{v^2}{r} = \dots = 2.92 \times 10^{-4} \text{ N}$$

- (b) What is the minimum value of the coefficient of static friction between the fly and the vinyl record required to prevent the fly from sliding off? (2)

2

$$f_{s, \max} \leq \mu_{s, \min} \times mg$$

$$\mu_{s, \min} \geq \frac{f_{s, \max}}{mg} = \dots$$

$$= 0.149 \checkmark$$

FBD

END OF TEST

